Jaggala Abhinauv Reddy

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.57
Sri Chaitanya Junior Kalasala, West Marredpally	Class XII (TGBIE)	2024	98.30%
St. Andrews High School, Suchitra	Class X (CBSE)	2022	98.60%

SKILLS

Programming Languages: C++, Java(Elementary), Python, SQL

Tools: Git, GitHub, Jupyter Notebook, Streamlit

LLM / GenAI: Ollama, Gemma, Prompt Engineering

Supervised Learning: Linear Regression, Random Forest Regressor, KNN, Gradient Boost, XGBoost

Data Analysis: NumPy, Pandas, Matplotlib, MS Excel

Data Structures: Arrays, Vectors, Linked Lists, Stacks, Queues, Binary Trees, Binary Search Trees, Graphs

PROJECTS

Quiz Generator (Local LLM):

- Built a local LLM-based quiz generator using **Ollama and Gemma** to create topic-specific multiple-choice quizzes directly on the user's system.
- Developed a **prompt-driven quiz generation pipeline** with options to set the topic, number of questions, and difficulty level.
- Structured the generated output to include **questions, multiple-choice options, correct answers, and explanations** for each quiz.
- Created an interactive **Streamlit interface** to generate quizzes, attempt questions, and view scores with answer explanations.

Concrete Performance Modelling:

- Built a **concrete compressive strength prediction model** using the UCI Concrete Compressive Strength dataset, applying machine learning regression techniques to predict strength from mix composition and curing age.
- Performed **data preprocessing and exploratory data analysis** including feature inspection, missing-value checks, statistical analysis, correlation analysis, and train-test splitting.
- Trained and evaluated **Linear Regression, Random Forest, Gradient Boosting and XGBoost** models using metrics such as **R^2, MAE, and RMSE**, selecting the best-performing model based on validation performance.
- Modularized the project into reusable Python components for **data preprocessing, model training, prediction, and model persistence**, using Joblib for saving trained model artifacts.

Expense Tracker: GitHub

- Developed a **Python-MySQL expense tracker** with **CRUD operations** to manage personal expenses through a **CLI**, supporting predefined categories, custom entries, date handling, and input validation.
- Implemented **SQL aggregation queries** to generate monthly, category-wise, and overall spending summaries, enabling efficient analysis of expense trends, budgeting patterns, and spending distribution across categories.
- Built **search**, **edit**, and **delete** functionalities to efficiently retrieve, update, and manage expense records, ensuring **data consistency**, improving usability, and simplifying day-to-day expense tracking.
- Structured the application into **modular Python components** by separating database operations, business logic, and user interaction, improving code organization, maintainability, and readability.

POSITIONS OF RESPONSIBILITY

Head, Operations and Logistics -- TEDx IIT Hyderabad

- Leading end-to-end **operational planning and strategy** for TEDxIITHyderabad 11th edition, supervising cross-functional execution, resource allocation, and workflow management.
- **Streamlining vendor management and event logistics** to optimize on-ground execution and resource efficiency across major event editions.

Operations Coordinator -- TEDx IIT Hyderabad

- Managed **venue logistics, equipment handling, crowd control, and real-time scheduling** for over **900** attendees during the **10th edition** workshop event.
- Supervised **volunteer teams and live on-ground operations**, coordinating speaker logistics and hospitality workflows to ensure seamless event execution.

Coordinator - EBSB IITHyderabad:

- Coordinated **venue, AV, vendor transit, and equipment logistics** across 6 EBSB cultural events, ensuring 100% on-time execution.
- Led **floor and stage crews**, resolving live technical issues and managing inventory to deliver 6 seamless, campus-wide showcases with zero downtime.

EXTRACURRICULAR

Tower Crane Competition:

- Designed and built a model tower crane using popsicle sticks and glue for the Civil Engineering Society's competition.
- Applied load distribution and truss stability principles to optimize the crane's lifting capacity and frame rigidity.